Detection Lab v1: Attack Detection Using Linux Logs

Author: Ashan Sandika

Lab Environment: Kali Linux & Metasploitable

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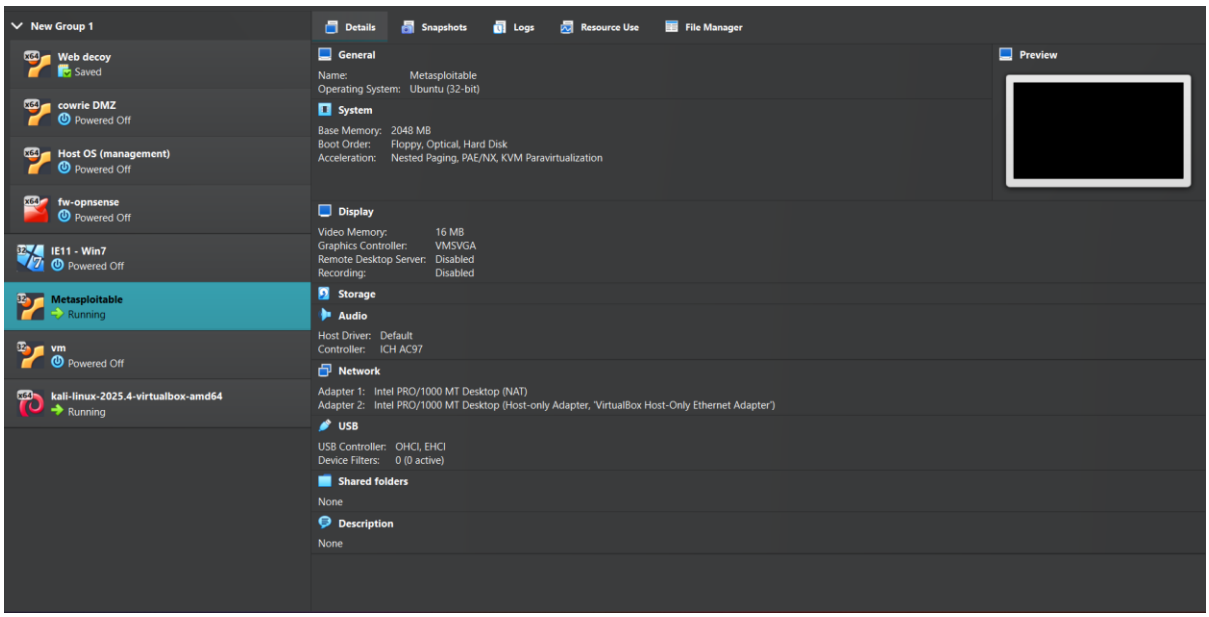
1. Introduction

This project demonstrates a basic detection engineering lab using Kali Linux as the attacker and Metasploitable as the vulnerable Linux target. The objective was to generate attack activity and detect it using system logs and a custom Bash-based detection script.

2. Lab Setup

Virtual Machine	IP Address	Role
Kali Linux	192.168.56.100	Attacker
Metasploitable	192.168.56.101	Target + Log Source



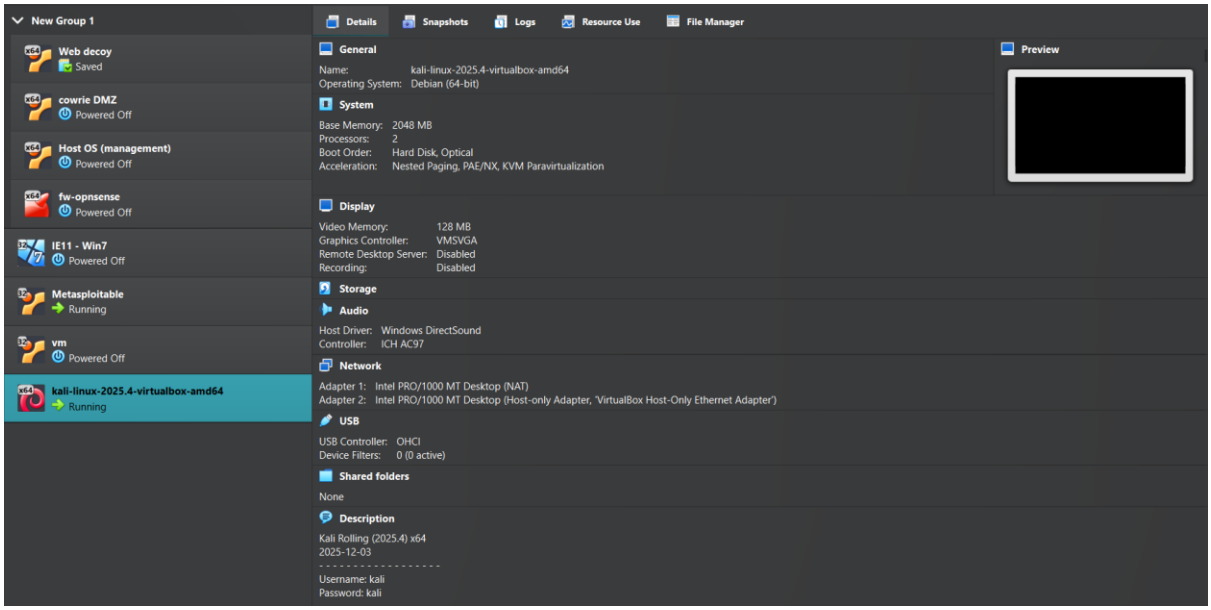


Figure 1: Lab setup showing attacker and target machines in a host-only network.

3. Attack Phase — Nmap Scan

A reconnaissance scan was performed from the Kali Linux machine to identify active hosts and services within the network.

Command used: nmap 192.168.56.0/24

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.56.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 04:58 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0018s latency).
All 1000 scanned ports on 192.168.56.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 0A:00:27:00:00:15 (Unknown)

Nmap scan report for 192.168.56.101
Host is up (0.00074s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:CB:FB:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.56.100
Host is up (0.0000020s latency).
All 1000 scanned ports on 192.168.56.100 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 256 IP addresses (3 hosts up) scanned in 19.33 seconds
```

Figure 2: Nmap scan identifying the target system at 192.168.56.101.

4. Log Evidence (Syslog Analysis)

After performing the scan, the Metasploitable system generated multiple log entries indicating incoming connections from the attacker IP address.

Command used: `cat /var/log/syslog | grep 192.168.56.100`

Observation:
Multiple "connect from 192.168.56.100" entries were recorded.

```
msfadmin@metasploitable:~$ cat /var/log/syslog | grep 192.168.56.100
Apr 20 10:12:08 metasploitable in.telnetd[7867]: connect from 192.168.56.100 (19
2.168.56.100)
Apr 20 10:12:08 metasploitable in.rshd[7869]: connect from 192.168.56.100 (192.1
68.56.100)
Apr 20 10:12:08 metasploitable postfix/smtpd[7863]: connect from unknown[192.168
.56.100]
Apr 20 10:12:08 metasploitable postfix/smtpd[7863]: lost connection after CONNEC
T from unknown[192.168.56.100]
Apr 20 10:12:08 metasploitable postfix/smtpd[7863]: disconnect from unknown[192.
168.56.100]
Apr 20 10:12:08 metasploitable in.rlogind[7871]: connect from 192.168.56.100 (19
2.168.56.100)
Apr 20 10:12:08 metasploitable in.rexecd[7872]: connect from 192.168.56.100 (192
.168.56.100)
Apr 20 10:12:14 metasploitable in.rlogind[7886]: connect from 192.168.56.100 (19
2.168.56.100)
Apr 20 10:12:20 metasploitable in.rshd[7891]: connect from 192.168.56.100 (192.1
68.56.100)
Apr 20 10:15:28 metasploitable postfix/anvil[7870]: statistics: max connection r
ate 1/60s for (smtp:192.168.56.100) at Apr 20 10:12:08
Apr 20 10:15:28 metasploitable postfix/anvil[7870]: statistics: max connection c
ount 1 for (smtp:192.168.56.100) at Apr 20 10:12:08
```

Figure 3: Syslog entries showing multiple connections from attacker IP.

5. Detection Rule Design

A simple detection rule was designed to identify scanning behavior:

- If a single IP generates **10 or more connection attempts**, trigger an alert.

This rule simulates basic reconnaissance detection used in real Security Operations Centers (SOC).

6. Custom Detection Script

A Bash script was developed to automate detection of:

- Multiple failed login attempts
- Root login activity
- Scan-like behavior

Detection Script:

```
#!/bin/bash

AUTHLOG="/var/log/auth.log"
SYSLOG="/var/log/syslog"
ALERT="/home/msfadmin/alerts.log"
IP="192.168.56.100"

echo "----- Detection Run: $(date) -----" >> "$ALERT"

FAILED_COUNT=$(grep "Failed password" "$AUTHLOG" | wc -l)

if [ "$FAILED_COUNT" -ge 5 ]; then
   echo "[ALERT] Multiple failed logins detected: $FAILED_COUNT attempts" >> "$ALERT"
fi

ROOT_LOGIN=$(grep "Accepted password for root" "$AUTHLOG")

if [ -n "$ROOT_LOGIN" ]; then
   echo "[HIGH ALERT] Root login detected!" >> "$ALERT"
fi

SCAN_COUNT=$(grep "$IP" "$SYSLOG" | wc -l)
```

```
if [ "$SCAN_COUNT" -ge 10 ]; then
    echo "[ALERT] Possible scan detected from $IP: $SCAN_COUNT connections" >> "$ALERT"
fi
```

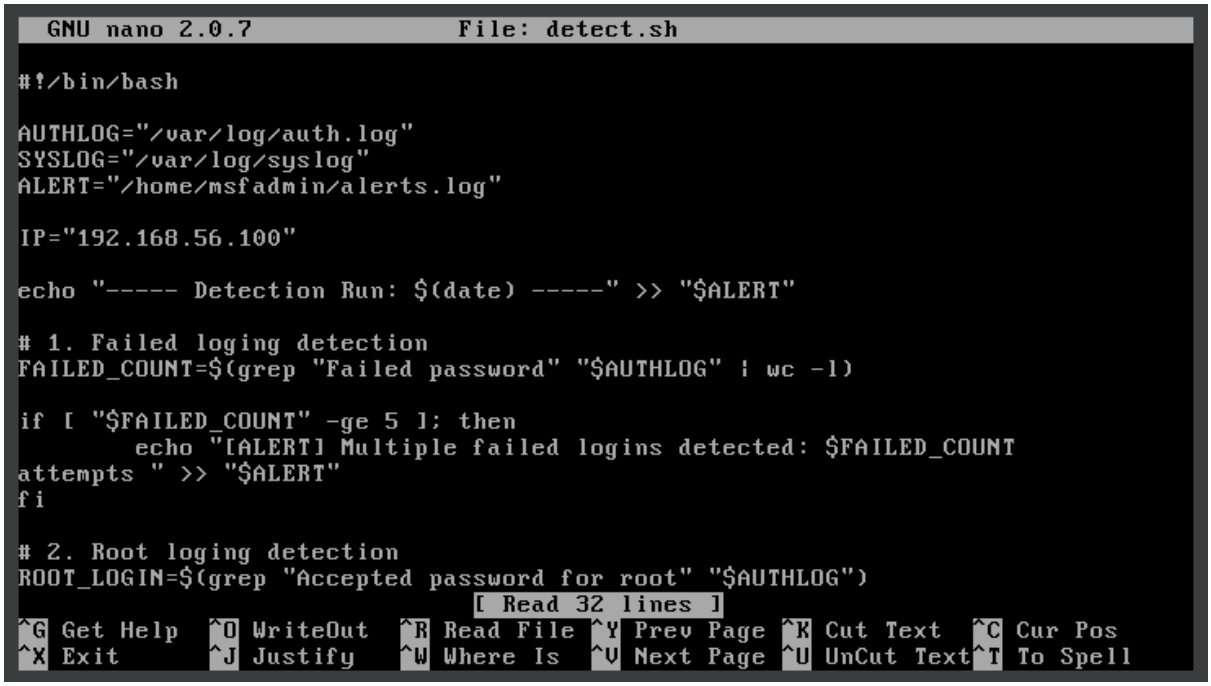


Figure 4: Custom detection script implementation.

7. Script Execution

The script was executed to analyze logs and generate alerts.

Commands used:

```
chmod +x detect.sh
./detect.sh
cat alerts.log
```

```
----- Detection Run: Fri Apr 24 01:20:00 EDT 2026 -----
[ALERT] Possible Scan detected from 192.168.56.100: 11
connections
----- Detection Run: Fri Apr 24 01:20:03 EDT 2026 -----
[ALERT] Possible Scan detected from 192.168.56.100: 11
connections
----- Detection Run: Fri Apr 24 01:20:04 EDT 2026 -----
[ALERT] Possible Scan detected from 192.168.56.100: 11
connections
----- Detection Run: Fri Apr 24 01:20:06 EDT 2026 -----
[ALERT] Possible Scan detected from 192.168.56.100: 11
connections
----- Detection Run: Fri Apr 24 01:20:59 EDT 2026 -----
[ALERT] Possible Scan detected from 192.168.56.100: 11
connections
----- Detection Run: Fri Apr 24 01:22:27 EDT 2026 -----
[ALERT] Possible Scan detected from 192.168.56.100: 11
connections
----- Detection Run: Fri Apr 24 03:23:31 EDT 2026 -----
[ALERT] Possible Scan detected from 192.168.56.100: 11
connections
----- Detection Run: Fri Apr 24 03:23:47 EDT 2026 -----
[ALERT] Possible Scan detected from 192.168.56.100: 11
connections
msfadmin@metasploitable:~$ _
```

Figure 5: Alert output generated after script execution.

8. Detection Results

The detection script successfully identified reconnaissance activity originating from the attacker machine.

Sample Alert Output:

[ALERT] Possible scan detected from 192.168.56.100

This confirms that multiple connection attempts were detected and flagged correctly.

9. Conclusion

This lab successfully demonstrated a basic detection engineering workflow. Attack activity was generated using Kali Linux and analyzed on the Metasploitable system using system logs. A custom detection script was developed to identify suspicious behavior such as scanning and authentication anomalies.

This project simulates a simplified Security Operations Center (SOC) environment and demonstrates foundational blue team skills in log analysis and detection.